OpenGL Cheat Sheet

OpenGL 4.6 / GLSL 4.60 · OpenGL ES 3.2 / GLSL ES 3.20 · EGL 1.5 · checked 1 November 2026 · Irurueta Docs, Graphics & Games

API family and versions

API	Latest	Notes
Desktop GL	4.6 (2017; spec rev. 2022-05-05)	core and compatibility; no 4.7 or 5.0
GLSL	4.60 (2023-08-14)	SPIR-V core in 4.6 (GL_ARB_gl_spirv)
OpenGL ES	3.2 / GLSL ES 3.20	ES 1.x deprecated
EGL	1.5	display, config, surface, context
WebGL	1 = ES 2.0, 2 = ES 3.0	ANGLE is the default backend on Windows in Chrome and Firefox
macOS	4.1 core cap	GL deprecated 10.14, ES on iOS 12 (2018)
Android	ES 2.0 API 8+, 3.0 API 18+, 3.1 API 21+, 3.2 API 24+	ANGLE optional from Android 15, manifest request from 17; Vulkan advised for new work
Layers	ANGLE, Zink	ANGLE: GLES on D3D11, GL, Vulkan, Metal. Zink: GL 4.6 on Vulkan
Extension	GL_EXT_mesh_shader	registered October 2025

Core 3.1+ has no fixed function. Extensions: KHR (ratified), ARB (board approved), EXT (multi-vendor), NV/AMD/INTEL (one vendor).

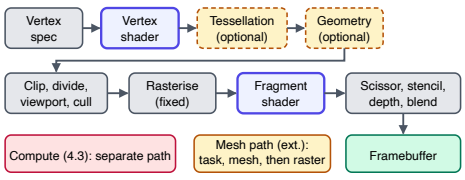
Context setup (GLFW + glad2)

```
glfwInit();
glfwWindowHint(GLFW_CONTEXT_VERSION_MAJOR, 4);
glfwWindowHint(GLFW_CONTEXT_VERSION_MINOR, 1); // 6 off macOS
glfwWindowHint(GLFW_OPENGL_PROFILE,
               GLFW_OPENGL_CORE_PROFILE);
glfwWindowHint(GLFW_OPENGL_FORWARD_COMPAT, 1); // macOS
glfwWindowHint(GLFW_OPENGL_DEBUG_CONTEXT, 1);
win = glfwCreateWindow(w, h, "t", 0, 0);
glfwMakeContextCurrent(win);
gladLoadGL(glfwGetProcAddress);
glfwSwapInterval(1);                  // vsync
glEnable(GL_DEBUG_OUTPUT);            // 4.3+, ES 3.2
glDebugMessageCallback(cb, 0);
while (!glfwWindowShouldClose(win)) {
  glClear(GL_COLOR_BUFFER_BIT |
GL_DEPTH_BUFFER_BIT);
  /* draw */ glfwSwapBuffers(win); glfwPollEvents();
}
```

Object lifecycle: create, bind or DSA, data, use, delete

Object	Create	Bind / data	Delete
Buffer	glGenBuffers (DSA glCreateBuffers)	glBindBuffer, glBufferData or glBufferStorage (DSA glNamedBufferStorage)	glDeleteBuffers
VAO	glGenVertexArrays (glCreateVertexArrays)	glBindVertexArray, attribute setup, element buffer	glDeleteVertexArrays
Texture	glGenTextures (glCreateTextures)	glActiveTexture + glBindTexture (glBindTextureUnit), glTexStorage2D, glTexSubImage2D	glDeleteTextures
Sampler	glGenSamplers	glSamplerParameteri, glBindSampler(unit)	glDeleteSamplers
FBO	glGenFramebuffers (glCreateFramebuffers)	glBindFramebuffer, glFramebufferTexture2D, glCheckFramebufferStatus	glDeleteFramebuffers
Program	glCreateShader, glCreateProgram	glShaderSource, glCompileShader, glAttachShader, glLinkProgram, glUseProgram	glDeleteProgram

Pipeline at a glance



Minimal shader pair, GL vs ES headers

```
// GL: #version 410 core    ES: #version 300 es
// ES fragment also needs: precision mediump float;
// vertex shader
layout(location=0) in vec3 aPos;
layout(location=1) in vec2 aUV;
uniform mat4 uMVP;
out vec2 vUV;
void main(){ vUV = aUV;
  gl_Position = uMVP * vec4(aPos, 1.0); }
// fragment shader
in vec2 vUV;
uniform sampler2D uTex;
out vec4 oColor;
void main(){ oColor = texture(uTex, vUV); }
```

Precision (ES): `lowp`, `mediump`, `highp`; vertex default highp, fragment none. `binding` qualifier needs GLSL 4.20 / ES 3.10.

Vertex attributes: stride and offset




```
glVertexAttribPointer(0,3,GL_FLOAT,GL_FALSE,20,
(void*)0);
glVertexAttribPointer(1,2,GL_FLOAT,GL_FALSE,20,
(void*)12);
glEnableVertexAttribArray(0);
glEnableVertexAttribArray(1);
// 4.3+: separate format and binding
glVertexAttribFormat(1, 2, GL_FLOAT, GL_FALSE, 12);
glVertexAttribBinding(1, 0);
glBindVertexBuffer(0, vbo, 0, 20);
```

MVP chain, NDC and depth

`clip = P * V * M * vec4(pos,1)`; divide by `w` gives NDC, then `glViewport` and `glDepthRange` give window coordinates.

```
M = glm::translate/rotate/scale(...)
V = glm::lookAt(eye, center, up)
P = glm::perspective(fovy, aspect, n, f)
glUniformMatrix4fv(loc,1,GL_FALSE,glm::value_ptr(mvp
));
```

- Matrices are column-major; view space is right-handed, looking down -Z.
- NDC x, y, z are in -1..1 (window origin lower left); Vulkan, Metal, D3D use depth 0..1.
- `glClipControl(GL_LOWER_LEFT, GL_ZERO_TO_ONE)` (4.5) with `glClearDepth(0)` and `GL_GREATER` gives reverse-Z; use `GLM_FORCE_DEPTH_ZERO_TO_ONE`.
- Resize: update `glViewport` and the aspect ratio.

Textures and samplers

Setting	Values
Storage	`glTexStorage2D(tgt, levels, fmt, w, h)`; levels = floor(log2(max(w,h)))+1
Min filter	`NEAREST`, `LINEAR`, `*_MIPMAP_NEAREST`, `*_MIPMAP_LINEAR`
Mag filter	`NEAREST`, `LINEAR`
Wrap S, T, R	`REPEAT`, `MIRRORED_REPEAT`, `CLAMP_TO_EDGE`, `CLAMP_TO_BORDER`
Mipmaps	`glGenerateMipmap(tgt)`; a mip min filter without a complete chain samples black
Anisotropy	`GL_TEXTURE_MAX_ANISOTROPY` (core in 4.6)
sRGB	`GL_SRGB8_ALPHA8` for colour maps; linear formats for normals
Compressed	BCn desktop; ETC2 (ES 3.0, GL 4.3); ASTC (ES 3.2 or extension); KTX2 and Basis to transcode
Upload	`glPixelStorei(GL_UNPACK_ALIGNMENT, 1)` for odd RGB widths

State toggles

Test	Calls
Depth	`glEnable(GL_DEPTH_TEST)`, `glDepthFunc(GL_LESS)`, `glDepthMask`
Cull	`glEnable(GL_CULL_FACE)`, `glCullFace(GL_BACK)`, `glFrontFace(GL_CCW)`
Blend	`glEnable(GL_BLEND)`, `glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA)`; premultiplied: `GL_ONE, GL_ONE_MINUS_SRC_ALPHA`; `glBlendEquation`
Stencil	`glEnable(GL_STENCIL_TEST)`, `glStencilFunc(f,ref,mask)`, `glStencilOp(sf,df,dp)`

Draw transparent surfaces last, sorted back to front, with depth writes off.

Draw-call family

Call	Use
`glDrawArrays(mode, first, count)`	unindexed
`glDrawElements(mode, count, type, offset)`	indexed
`glDrawElementsInstanced(..., n)`	instancing, divisor per attribute
`glMultiDrawElementsIndirect`	GPU-written commands (4.3)
`glDispatchCompute(x, y, z)`	compute (4.3, ES 3.1)
`glDrawMeshTasksEXT(x, y, z)`	task and mesh (extension)

std140 vs std430

Type	std140 (UBO, SSBO)	std430 (SSBO)
scalar	4	4
vec2	8	8
vec3, vec4	16	16
array element	rounded up to 16	base alignment, tightly packed
struct	aligned and sized to a multiple of 16	largest member alignment
mat4	4 vec4 columns	4 vec4 columns

Binding: `glBindBufferBase(GL_UNIFORM_BUFFER, i, buf)`; `layout(std140, binding=i)` needs 4.20, else `glUniformBlockBinding`. SSBO and std430 need 4.3 or ES 3.1. A `vec3` aligns to 16 bytes, and in std140 a `float` array uses 16 bytes per element.

Debugging checklist

- `glGetError()`: polled, reports one flag, does not say which call; use only as a fallback.
- `KHR_debug` (core 4.3, ES 3.2): `glDebugMessageCallback`, `glObjectLabel`, `glPushDebugGroup`.
- Capture frames with RenderDoc 1.47 (GL and ES) or apitrace.
- Black output: no VAO or program bound, uniform optimised out (location -1), incomplete texture or framebuffer, depth not cleared, wrong stride or offset, culled by winding.
- Check `glCheckFramebufferStatus`, shader and link logs after every build.

Porting map: GL to Vulkan, Metal and WebGPU

GL	Vulkan	Metal	WebGPU
context	instance, device, queue	MTLDevice	GPUDevice
buffer	VkBuffer + memory	MTLBuffer	GPUBuffer
texture, sampler	VkImage, VkSampler	MTLTexture	GPUTexture
GLSL	SPIR-V	MSL	WGSL
state calls, VAO	VkPipeline	render pipeline state	GPURenderPipeline
uniforms, binding points	descriptor sets	argument tables	bind groups
immediate calls	command buffers	command encoders	command encoders
implicit sync	barriers, fences	fences, events	queue order

Changed since older tutorials

- `glBegin`/`glEnd`, matrix stack, fixed-function lighting: gone from core in 3.1 and 3.2 (2009); use shaders, VAOs and GLM.
- GLUT is unmaintained: use GLFW or SDL with glad2.
- "GL 2.1 is the latest": it is 4.6; GLSL 1.10 and 1.20: use `#version 410 core` or 460.
- Selection mode picking: use an ID buffer or a ray.
- ES 2.0 only: ES 3.0 to 3.2 are universal on current Android.
- Eclipse and ADT: Android Studio and Gradle.
- AGL and NSOpenGL: OpenGL is deprecated on Apple platforms; use Metal.